Fusing LLM Semantic Repair with Genetic Programming for Robust Alpha Discovery

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*Vanilla quant models break out-of-sample. A regime-aware GP + local-LLM repair loop fixes that — S&P 500 Sharpe decay cut from −35% **to** −7%, China's from −92% to −47%, and no candidate tree is ever thrown away — the whole loop costs one monthly pass on a single workstation.*

10.6% net return (peak) | **0.65** net Sharpe | **−7%** S&P decay (was −35%) | **+18%** over quant ceiling

NSE Nifty 50 2002–2025 (23 yrs) ▪ validated on S&P 500 + SSE ▪ costs 20+10 bps, cap INR 50 Cr ▪ local Llama-3.2-3B + Gemma-2-2B, $0 inference ▪ one monthly rebalance, judged and repaired in minutes

0 Objectives

- **Primary:** discover symbolic ranking formulas that maximise forward risk-adjusted return across regimes while cutting out-of-sample decay.
- **Secondary:** keep every candidate via repair-not-reject; stay cheap (local 3B/2B, $0) and tradable (costs + cap priced in).
- **Validate:** NSE Nifty 50 2002–2025, transfer to S&P 500 + SSE with one fixed setup.

1 Problem in 30 seconds

- Quant factors **decay**: top-1000 anomalies lose about 32% after publication; GP memorises noise.
- Old LLM+GP hybrids **delete** bad formulas and lose diversity.
- Fix: **repair, don't reject** — rewrite only the truly nonsensical (score 2 or less).
- Deletion is expensive: a dropped tree takes its partial signal with it; a thinner pool overfits one lucky regime.
- Local 3B/2B models make the judge cheap: $0 API cost, prompts stay on-workstation, latency fits monthly rebalance.
- Claim stays narrow: repair lifts out-of-sample stability, not in-sample fit; risk budget never widens.

2 What each layer does

- **AFM (slow):** news mood + PE/ROE/EPS/BV gives a bull/bear prior. It drifts over weeks, so one noisy headline never flips it.
- **HMM (fast):** 2-state volatility filter: calm to trend, turbulence to reversals + quality (Baum–Welch + Viterbi, daily).
- **GP (inventor):** 500 trees, depth max 6, Sharpe-minus-complexity fitness; bloat is charged from generation one.
- **LLM (editor):** Llama rates economics 1–5; Gemma rewrites failures from a closed vocabulary ($L \leq 5$ typed edits max).
- **Guards:** repaired trees must pass AST + validation ranges before re-entering; failures never replace parents.
- **Settings:** FinBERT $\tau = 0.5$; GP pop 500, depth 6; dollar-neutral top/bottom-15%, monthly rebalance; identical on Nifty, S&P, SSE.

4 Decay cut in half (Sharpe)

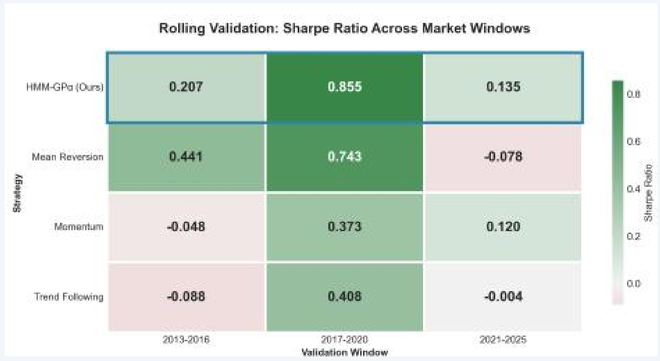
Out-of-sample Sharpe vs the in-sample peak (3 engines/market): structure that survives the test window bleeds less. Conditioning helps, repair helps more.

Market / engine	OOS Sharpe	Decay
S&P Vanilla GP	0.905	-35.03%
S&P HMM-GP	0.974	-20.71%
S&P **LLM-GP (ours)**	**1.075**	-6.77%
SSE Vanilla GP	0.035	-92.08%
SSE HMM-GP	0.232	-63.28%
SSE **LLM-GP (ours)**	**0.067**	-46.68%

S&P decay -35.03% to -6.77%; China -92.08% to -46.68%. Vanilla GP overfits; conditioning removes part, repair removes the rest. SSE level (0.067) stays small, so the claim there is halved decay, not a large Sharpe.

2b Loop + stability (how it runs, why it holds)

GP loop: AFM prior + HMM state → fitness (Sharpe − complexity) → Llama judge (1–5) → Gemma repair ($S \leq 2$, closed vocab, $L \leq 5$) → recompile, back into evolution. Trees scoring 3–5 pass untouched, so the editor spends compute only where the economics is broken. Zero attrition: every repair compiles, the pool never drops below its 500 trees, and nothing is discarded for looking unusual — only for scoring low.



Rolling-window Sharpe stays positive (0.855) in turbulent 2017–20 where momentum + trend turn negative. Regime specialists, not one average model: every rolling window is judged on its own regime, so a calm 2013–16 never has to average out a hostile 2017–20.

3 System architecture (paper Fig. 1, middle)

Left: mine candidates from data + regimes. Right: repair — never delete — what the judge rejects. Lower: fuse, size, execute.

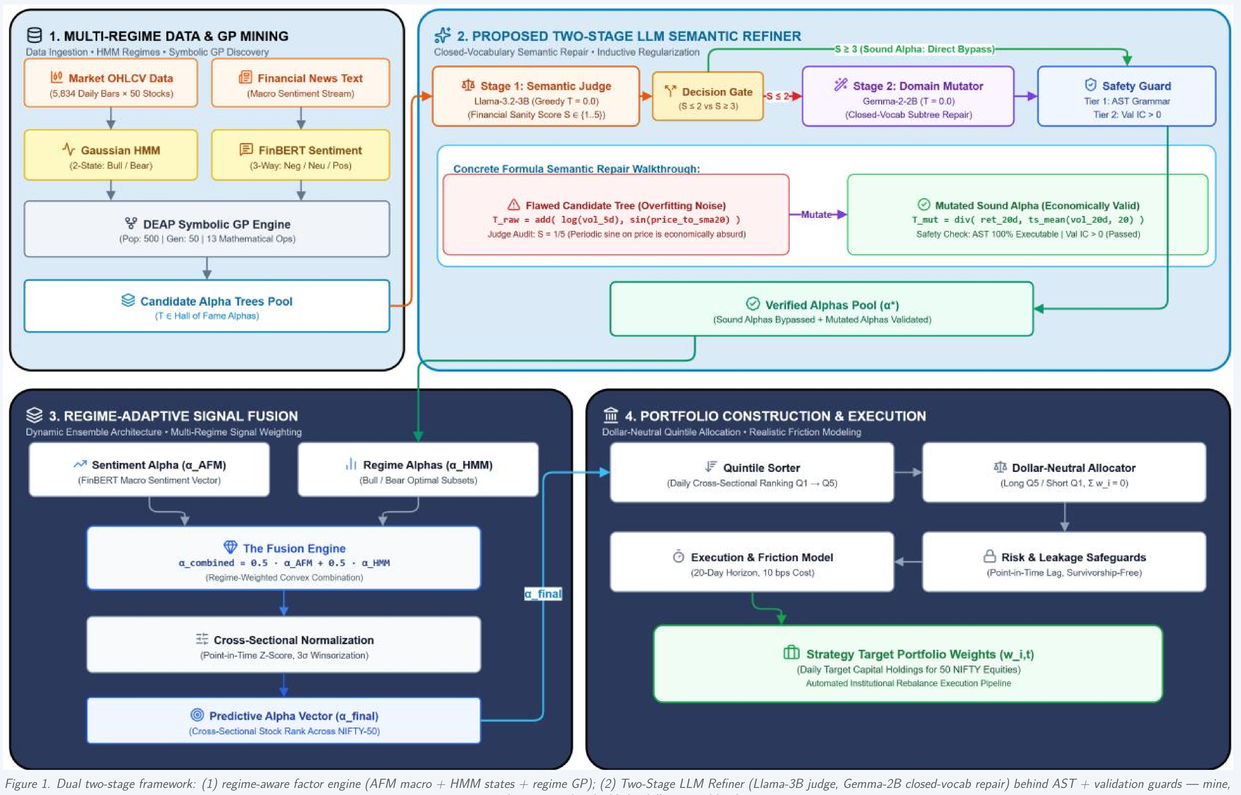
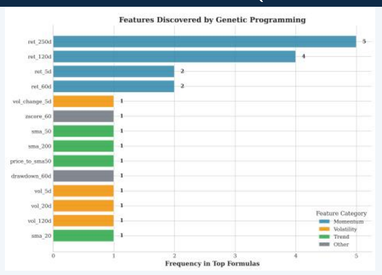


Figure 1. Dual two-stage framework: (1) regime-aware factor engine (AFM macro + HMM states + regime GP); (2) Two-Stage LLM Refiner (Llama-3B judge, Gemma-2B closed-vocab repair) behind AST + validation guards — mine, judge, repair, then build the dollar-neutral book.

5 What the machine rediscovers (feature mix)



Feature mix of top alphas (paper Fig. 6): intermediate momentum + volatility scalers dominate. Each bar counts appearances of that feature in the top-ranked formulas: a trend term you can hold in calm regimes plus a scaler that trims size when vol rises, with reversals + quality for stress. The categories stay classical (trend, momentum, volatility, quality), so every repaired formula still reads like something a desk would write. Because the inputs stay classical, the repair step can never smuggle in data the GP never saw.

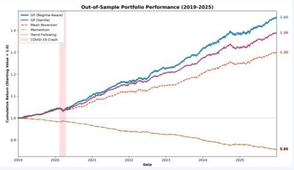
6 Read this and leave (5-min summary)

- **Robust every year:** positive Sharpe 2019–25 incl. COVID (-14.8% DD vs -38.5% momentum).
- **Deployable:** about $250M capacity; 20+10 bps costs priced in, INR 50 Cr cap.
- **Cheap to run:** one monthly pass on a single workstation — local 3B/2B models, no cloud API, $0 inference.
- **Where it wins:** decay control — S&P keeps 93% of its backtest Sharpe, SSE keeps about half instead of a tenth.
- **Honest limits:** the SSE level (0.067) is still small, and transfer is tested on one US plus one Chinese index.
- **Next:** NSE 500, LoRA-Gemma, multi-agent judges.

Take-away: semantic repair is a free regulariser, India to US to China, at zero inference cost.

7 Repair example + wealth curve + regimes (one glance)

Before (1/5): `add(log(vol_5d), sin(p_to_sma20))` — sine on prices is noise: it oscillates with no economic channel, so it fits one window and dies in the next. After (4/5): `div(ret_20d, ts_mean(vol_20d,20))` — return scaled by its own trailing volatility: risk-adjusted momentum, the line a quant desk would write by hand. Rank IC −0.012 → +0.038; 100% repairs compile, zero crashes over 1,729 days. Both edits stay inside the original operator set, so no new primitive has to be back-tested.



Wealth 2019–25 compounds through bull + bear while a plain momentum book gives back −40.65% at its worst.



HMM: State 0 calm expansion vs State 1 turbulence (COVID-2020, tightening-2022); stickiness 0.994/0.983 — the model rarely flips states, so the routing signal stays stable.

Repair turns nonsense into economics; regime routing is why wealth survives shocks. Read together: cleaner operators on top, regime-aware routing below, one smooth equity curve at the end.

Refs: Kakushadze 101 Alphas (2016) ▪ AlphaAgent arXiv:2502.16789 ▪ TEDE (2026) ▪ DEAP JMLR ▪ Hamilton (1989) McLean (2016) ▪ Almgren–Chriss (2000). NSE 50 2002–25; S&P 500 + SSE transfer test. Draft 1 (poster.pdf) is the full version behind every number shown above.